Component-Level Energy Benchmarking of Reasoning in LLMs

Course: Programming AI Accelerators
Team: 3 (Mehul Ke Dost Part 2) — Aaditya Bhatia, Nidhish Jain, Virat Garg
Model category: Language Models (reasoning LLMs, math + coding benchmarks) — DeepSeek-R1-Distill-Qwen (Qwen2.5-based) for component analysis, Qwen3 for a native thinking-on/off comparison
Hardware: Inference only, no training

The Idea

Reasoning ("thinking") models spend extra inference-time compute generating a chain-of-thought before answering. Two independent measurement studies — Jin, Wei & Brooks (Harvard, arXiv:2505.14733) and Oviedo et al. (Microsoft Research, *Joule* 2026, arXiv:2509.20241) — show this costs dramatically more energy: up to 113× in the worst case, and a consistent ~13× for typical reasoning-length outputs, measured at the **whole-model** level. Neither looks inside the model to see which component — Attention, MLP, KV-cache — is actually responsible. CLEAR (Jain et al., IIIT Hyderabad, arXiv:2510.02810), the only framework that does component-level energy attribution, has only been tested on short, fixed-length inputs, never on a real, growing reasoning trace. **The gap we're closing:** apply CLEAR's component-level measurement directly to reasoning traces, to see where reasoning's energy actually goes, whether it's spent productively, and whether cutting it short is a good trade.

Proposed Approach

We're building on CLEAR's own three-stage recipe (cache a component's activation, replay it many times to get past NVML's sampling noise, validate the result), not new instrumentation. Roughly, in the order we intend to attempt them:

- **Experiment A — does Attention take over as context grows?** From FLOP counting (MLP cost/token is ~constant under KV-caching, Attention cost/token grows with context length), there should be a point where Attention overtakes MLP as the dominant energy cost — and CLEAR's finding that Attention has disproportionate energy-per-FLOP suggests this shows up even more sharply in energy than in raw FLOPs. We checkpoint a long reasoning trace at intervals and measure component energy at each point — not certain yet how cleanly this shows up in practice.
- **Experiment B — where does energy go on wrong answers?** Jin et al. show incorrect traces run much longer than correct ones. We check, component-by-component, whether that wasted length is disproportionately an Attention cost or spread evenly.
- **Experiment C (our main experiment) — does cutting reasoning short help, and where?** Jin et al. propose but never test that truncating reasoning early could save ~14% energy for free. We implement this via forced early-stopping (inserting `</think>` before the model would naturally stop) at several token budgets, and measure accuracy plus *which* component's energy actually drops.
- A layer-by-layer (not just component-type) energy view is a possible bonus visualization if time allows.

Since DeepSeek-R1-Distill-Qwen has no native reasoning toggle, Experiment C's forced-truncation requires custom stopping logic. We additionally want to run the same thinking-on/off comparison natively on **Qwen3**, which supports a `/think`/`/no_think` switch and an official `thinking_budget` parameter that does this truncation for us — a useful lower-risk cross-check on the same phenomenon.

Baseline

Whole-model (non-component) energy and accuracy for reasoning vs. non-reasoning, measured before any component decomposition: reproducing Jin et al.'s Base-vs-RT numbers on DeepSeek-R1-Distill-Qwen, and independently on Qwen3 using its native toggle. This calibrates our measurement setup before trusting any component-level number built on top of it.

GPU Requirement

We require access to **at least one NVIDIA GPU with ≥24GB VRAM** (RTX 3090/4090, A5000, or L4-class) — sufficient to run the DeepSeek-R1-Distill-Qwen 7B tier and Qwen3 4B/8B with the long-context KV-cache growth our experiments need. A smaller 8–16GB GPU (RTX 3060/4060, Colab T4) works as a fallback restricted to the 1.5B / Qwen3-0.6B–1.7B tier only; ≥40GB (A100/A6000-class) would additionally let us test the 14B tier. No training is involved, only inference.

Software: Hugging Face `transformers` (not vLLM — CLEAR's method needs direct hook access to individual

modules), NVIDIA NVML for component-level energy, Zeus for whole-model sanity checks.

Evaluation Metrics

- Energy per query, whole-model and per-component (Joules/mJ)
- % Energy captured and measurement consistency (std. dev.) — CLEAR's own validation checks
- Energy per FLOP per component, vs. context length
- Task accuracy (exact-match/unit-test) at each truncation budget
- Output length, tracked alongside energy; fixed seeds and recorded run configs for every reported number

Expected Contribution

The first component-level (not whole-model) energy benchmark of reasoning traces: where Attention overtakes MLP as context grows, whether wasted energy on wrong answers is Attention-concentrated, and a validated, mechanistic answer to Jin et al.'s untested "cutting reasoning short saves energy" claim — cross-checked across two model families (DeepSeek-R1-Distill-Qwen and Qwen3).

Risks / Fallback Plan

- **GPU lands on a small-VRAM tier:** 1.5B/Qwen3-1.7B-only still supports all three experiments; only cross-model-size comparison is dropped.
- **Forced-truncation produces malformed answers at low budgets:** grade only budgets where a well-formed final answer is actually emitted; report the minimum viable budget as a finding.
- **CLEAR's validation thresholds (90% capture, <9.5% std. dev.) not reached:** report actual achieved numbers transparently; increase repetition count if noise-limited.
- **Time runs short:** priority is Experiment C > A > B; the Qwen3 cross-check and layer-heatmap visualization are cut first.

Timeline

- **Week 1:** Confirm GPU access, read core papers, set up environment.
- **Week 2:** Get CLEAR's measurement pipeline running end-to-end.
- **Week 3:** Reproduce the baseline numbers.
- **Week 4:** Run Experiment A, check whether the crossover shows up.
- **Week 5 — Eval 1:** Present baseline + Experiment A results.
- **Week 6:** Run Experiment C (truncation), on both model families.
- **Week 7 — Eval 2:** Present working A + C results; start Experiment B.
- **Week 8:** Finish Experiment B, finalize plots.
- **Week 9 — Final:** Full results, analysis, demo, report.

Division of responsibilities: all three members work across experiments rather than owning a single one in isolation, but broadly — Aaditya leads the CLEAR measurement pipeline and NVML/Zeus instrumentation, Nidhish leads running the experiments and data collection across both model families, and Virat leads analysis, plotting, and the report/presentation. Roles rotate across baseline, Experiment A, B, and C so everyone gets hands-on time with both the infrastructure and the results.

Related Work

- Jin, Wei, Brooks. *The Energy Cost of Reasoning.* arXiv:2505.14733
- Oviedo et al. *Energy Use of AI Inference, Efficiency Pathways, and Test-Time Scaling.* Joule 2026, arXiv:2509.20241
- Wilhelm, Wittkopp, Kao. *Beyond Test-Time Compute Strategies.* arXiv:2603.20224
- Jain, Goyal, Pandey, Vaidhyanathan. *Dissecting Transformers: A CLEAR Perspective Towards Green AI.* arXiv:2510.02810
- Chung et al. *Zeus.* NSDI 2023
- Chung et al. *The ML.ENERGY Benchmark.* NeurIPS D&B 2025, arXiv:2505.06371
- Chung, Wu, Ma, Chowdhury. *Where Do the Joules Go?* arXiv:2601.22076
- Niu et al. *TokenPowerBench.* AAAI 2026, arXiv:2512.03024
- Fadel Argerich, Fürst, Patiño-Martínez. *Watt Counts.* arXiv:2604.09048
- Vartziotis et al. *From Tokens to Watt-hours.* arXiv:2607.26571